

Problem Formation, Product Design and Specifications

Basic Terminology

- **Product:** It is something sold by an enterprise to its customers
- **Origin:** Product Design has evolved from using primitive tools and methods Centuries ago to using advanced technology and multi-disciplinary concepts in present age

Problem formation:

It is the important stage in Product Design

- **Steps involved:**
 - **Identify the user needs:** Gather insights and identify the major problems
 - **Research and Analysis:** Conduct market research to analyze the potential competitors and market conditions
 - **Problem statement:** Craft a clear and concise problem statement that clearly addresses the challenge
 - **Set clear goals:** Set clear objectives that the object aims to achieve and establish the Key performance indicators to assess the progress and success of the product

Tools and Techniques for Problem Formation

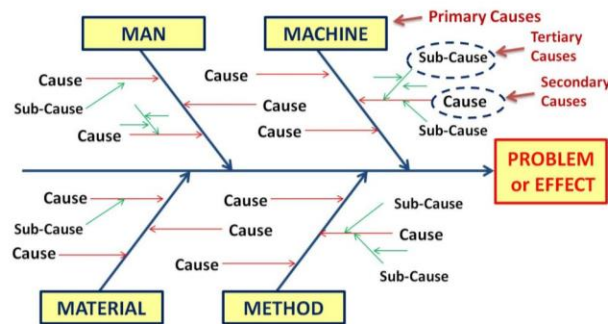
The various tools and techniques used for Problem Formation are summarized as follows:

S No	Tool/Technique	Function
1	5 why's method	Root Cause analysis
2	Fish bone diagram	Identify cause and effect
3	Stakeholder mapping	Understand stakeholder needs
4	Problem statement Canvas	Clearly articulate the problem
5	Jobs to be done	Understand user motivations
6	SWOT Analysis	Evaluate product factors
7	Mind Mapping	Organize ideas visually
8	User Personas	Define target users
9	Empathy Mapping	Understand user experience
10	Customer Journey Mapping	Track user interactions

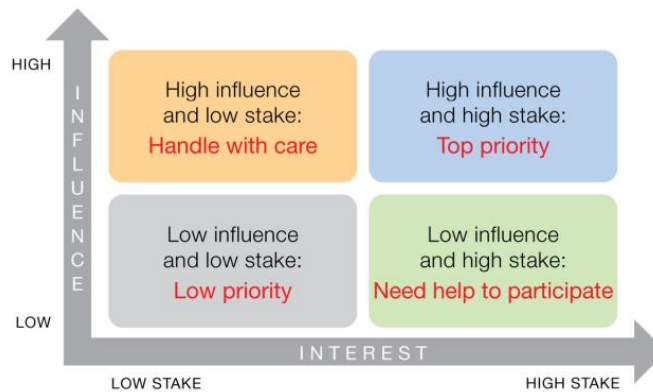
The techniques are detailed here:

1. **5 why's method:** This method allows asking why 5 times to get to the cause of the problem. This method ensures that the solution addresses the core issue, preventing superficial fixes that fail to resolve the actual problem. For example, someone is Late for the work. The series of responses to Why questions can be as follows - Woke up late, Alarm did not work, Battery exhausted, Forgot to check. The last response is the root cause of the problem.
2. **Cause and Effect Analysis using Fishbone Diagram:** This helps in visualizing potential causes of a specific problem. Head represents the problem and bones represent the categories of causes. By

mapping out these relationships, the diagram reveals the most likely root cause, making it easier to prioritize actions for resolution



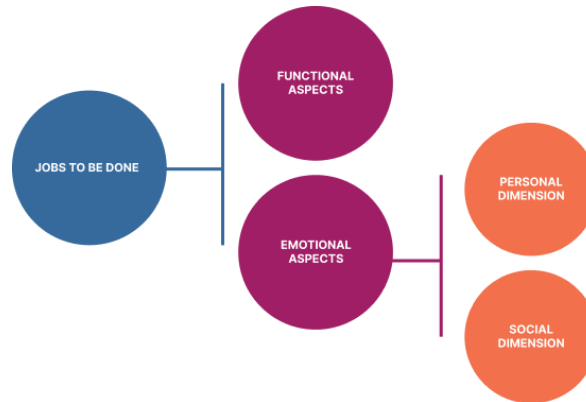
3. **Stakeholder Mapping:** It is a strategic tool used to identify and analyze the people, groups, or organizations that will be satisfied by or have an interest in the problem or product. This helps the design and problem solving process consider the perspectives of everyone involved from customers and users to internal teams and external partners. The relation among various elements is depicted in the Fig. below.



4. **Problem Statement Canvas:** It is a structured tool used to clearly define and breakdown a problem into essential components. It helps avoiding miscommunication, and focuses on team's efforts to develop a targeted solution.

CONTEXT When does the problem occur? <i>Every workday, in the mornings and evenings for an average of 2-3 hours per day</i>	PROBLEM What is the root cause of the problem? <i>Lose time in traffic instead of doing something more valuable</i>	ALTERNATIVES What do customers do now to fix the problem? <i>Sign up for Uber and accept rides only when going to or coming back from work</i>
CUSTOMERS Who has the problem most often? <i>Young men aged 25-35 with middle-low income, who live in suburban São Paulo and work in a corporate office in the city center</i>	EMOTIONAL IMPACT How does the customer feel? <i>Frustration & boredom</i> QUANTIFIABLE IMPACT What is the measurable impact (include units)? <i>Lose on average 40 hours per month</i>	ALTERNATIVE SHORTCOMINGS What are the disadvantages of the alternatives? <i>Driving for Uber requires more time waiting for a ride, as trip origin & destination might not coincide with their home-work itinerary</i>

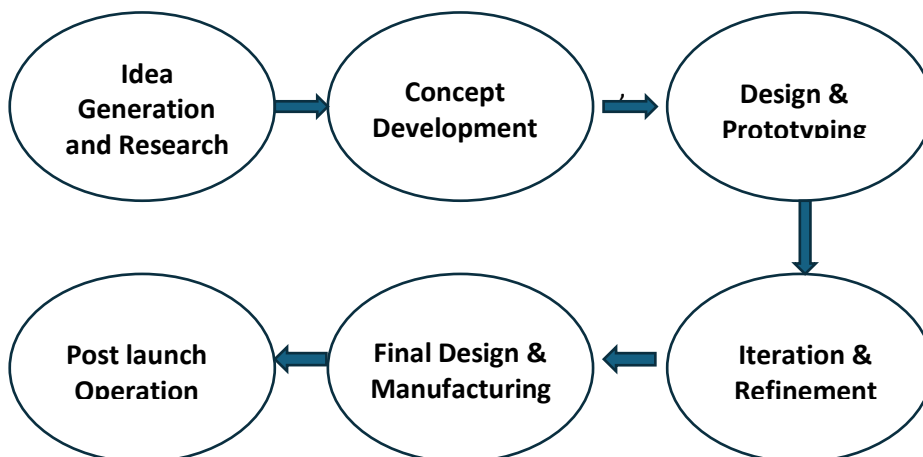
5. **Jobs to be done:** This approach focuses on understanding the tasks or jobs that users want to accomplish when they use a product or service. By focusing on functional, emotional or social needs, designers can uncover deeper motivations behind user behaviour, allowing them to create products that better align with what users are really trying to accomplish.



S No	Technique	Description
6	SWOT Analysis	Assess Strengths, Weaknesses, Opportunities and Threats related to the product or service
7	Mind Mapping	Create a visual representation of ideas and relationships to clarify the problem areas and encourage brainstorming
8	User Persona	Create fictional profiles representing different segments of users to ensure the problem focuses on their needs
9	Empathy Mapping	Visualize user's thoughts, feelings, and challenges to identify their pain points and unmet needs
10	Customer Journey Mapping	Map out the steps a user takes to interact with the product, identifying pain points and opportunities for improvement

Product Design

Product Design is the process of imagining, conceptualizing, testing, iterating and refining a product, service or experience until it is made ready for the user.



This requires a combination of research strategy, industry expertise, and creative problem solving. Coordinated effort is required from idea generation to experimentation and improvement till the final outcome. The product design involves the above steps. Refinement of product is conducted after taking initial feedback and the final design is improved. The approved final design is sent to manufacturing for generating multiple units. The product is launched and customer feedback is collected. The product undergoes its updates and iterations for refinement and performance of the product over its life time is targeted for improvement.

Key Factors in Product Design

A number of factors influence the Product Design. They are listed below.

Table: Key factors in Product Design

S No	Key Parameter	Description
1	Functionality	Product must fulfil its intended purpose efficiently and reliably
2	Aesthetics	Should have a good visual appeal and design to attract customers
3	Usability	Product must be simple and easy to use by everyone with minimal effort
4	Materials	High quality, sustainable materials should be used
5	Ergonomics	Product design should be comfortable easy to use and no long term effects on users
6	Sustainability	Product should be eco-friendly and should low environmental impact
7	Cost	Cost of the product should be affordable. Balancing the cost and quality of product is essential
8	Brand identity	Should reflect the brand's values and maintain the identity
9	Innovation	Incorporation of innovative technologies in the product is essential to meet increased needs
10	Compliance and Safety	Product design and its features should adhere to existing regulations. Safety standards should be maintained for its operation and build trust with users/customers

Product Strategies

These strategies outline a company's comprehensive approach to developing, marketing and selling its products. These strategies serve as a roadmap to achieving long term success and staying competitive in the ever-evolving marketplace. By incorporating these strategies, the producer can achieve efficient product growth, foster innovation, and build stronger customer relationships, and achieve long-term success in a competitive market place. A number of strategies are used by the industry which are outlined as follows:

Table: Product Strategies

S No	Strategy	Goal	Methods
1	Market Strategy	To gather insights into consumer needs, competitors and market trends to guide product development	Conduct primary research for fresh data, use secondary research to analyze information and study

Problem Formation, Product Design and Specifications

		and marketing decisions	competitors to identify market gaps
2	Target market Identification	To define specific audience segments most likely to benefit from and purchase a product	Segment markets based on demographics, psychographics (eg. Lifestyle, values) and behaviour to personalize products and marketing
3	Distribution Channels	To ensure products reach consumers in convenient locations and at right time	Choose between direct (online & offline sales) and indirect (retail partners, wholesalers) channels or a hybrid approach for broader reach and accessibility
4	Lifecycle Management	To maximize a product's profitability and market longevity by overseeing its journey through each lifecycle stage	Implement targeted strategies for each phase – create awareness, enhance reach, maintain interest in maturity, and adapt or phase out at decline
5	Product Differentiation Strategy	To distinguish a product by emphasizing unique features or benefits that set it apart from competitors	Focus on superior quality, design or customer service that adds distinctive values for consumers
6	Cost leadership Strategy	To be the lowest-cost producer in the industry, offering competitive pricing without sacrificing quality	Streamline the operations, utilize economies of scale, and optimize supply chain efficiencies
7	Market Segmentation strategy	To target specific customer segments with products that meet their unique needs	Conduct market research to identify segments and tailor products and marketing approaches accordingly
8	Innovation strategy	To lead the market through the continuous introduction of new or improved products	Invest in R&D, encourage creative thinking, and stay ahead of technological advances to meet evolving consumer needs
9	Customer focus strategy	To prioritize customer satisfaction by providing personalized products and services that meet individual needs	Gather and analyze customer feedback, offer customization options, and enhance the customer experience through exceptional service
10	Diversification strategy	To mitigate risk by expanding into new product categories or markets	Develop or acquire products in diverse categories or enter new markets to reduce dependence on a single revenue stream
11	Product line Extension strategy	To broaden the product line by offering variations such as new sizes or flavours, of existing products	Use market research to identify demand for variations and introduce new options under an established brand to attract different customer segments
12	Branding Strategy	To build a strong, recognizable brand that fosters customer loyalty and differentiates from competitors	Develop a clear brand identity, maintain consistent messaging, and create an emotional connection with

			consumers
13	Globalization strategy	To expand a product's reach into international markets	Adapt products to local tastes, comply with regulations, and establish a global distribution network
14	Digitalization strategy	To leverage digital technologies for enhanced product reach, customer engagement, and operational efficiency	Use digital platforms for e-commerce, data analytics for personalized marketing and CRM systems for streamlined customer service
15	Sustainability strategy	To promote environmental responsibility and reduce the ecological impact of produce and operations	Incorporate sustainable materials, reducing waste, adopting energy-efficient processes and ensuring ethical sourcing practices

Product Values

Product Value refers to the overall benefits and satisfaction a product or service provides to customers. It is a major aspect to attract and retain customers as it shapes their perceptions of worth. High Product Value meets or exceeds expectations, addressing needs and standing out from alternatives. Four different product values are available for practice which are discussed here as follows:

1. Functional Value:

- **Utility:** The core purpose of the product, assessing how well it performs its intended function
- **Quality:** Encompasses durability, reliability and overall performance
- **Features:** Additional functionalities that enhance this product's usefulness
- **Convenience:** Focuses on ease of use, accessibility and availability
- **Example:** Smart phones provide multiple functionalities like calls, internet access and apps, combined with durable materials and reliable performance

2. Emotional Value

- **Feelings & Experiences:** The emotional responses elicited by the product, such as pleasure & excitement
- **Brand Perception:** The emotional connection and trust customers develop with a brand
- **Aesthetics:** The design and visual appeal of the product that can evoke emotions
- **Example:** Luxury cars offers and exhilarating driving experience and convey status and prestige through their attractive designs

3. Social Value

- **Status & Identity:** The role of products in conveying social status or aligning with an individual's identity
- **Community & Belonging:** Products that foster connections within a group or community
- **Influence:** The ability of products to enhance social connections and influence
- **Example:** Designer clothing signifies status and identity, often aligning with special groups

4. Economic Value

- **Cost Savings:** The ability of a product to reduce expenses over time through efficiency or operational cost reduction
- **Return on Investment:** The financial benefits received relative to the cost of the product
- **Total Cost Ownership:** The complete cost of owning the product, including maintenance and operational expenses
- **Example:** Energy-efficient appliances may have a higher upfront cost but lead to significant savings on electricity bills over time

Importance of Product Values

- **Customer Satisfaction and Loyalty:** High perceived value leads to greater customer satisfaction, fostering loyalty and encouraging repeat purchases
- **Competitive advantage:** Understanding and enhancing product value allows businesses to differentiate themselves, whether through superior quality, customer service or unique branding strategies
- **Pricing strategy:** Recognizing the value offered helps businesses set competitive prices, maximizing profitability based on perceived customer willingness to pay
- **Market positioning:** Effective communication product value influences how a product is perceived in the market, attracting different consumer segments
- **Innovation and Development:** Understanding customer needs ensures product development, resulting in new products that enhance existing value propositions

Product Planning

- It is a crucial aspect of product management that encompasses a series of strategic steps aimed at determining what products to create or improve based on market needs
- It involves defining vision, features and requirements of a product while prioritizing these elements to best meet user needs and align with the company's strategic objectives

Objectives of Product Planning

- **Identify Market Needs:** Understanding the problems and desires of potential customers is fundamental to creating relevant products which involves what customers are looking for and how existing solutions fall short
- **Strategic alignment:** Product development must align with the overall business strategy and goals. Should ensure the new products support broader company objectives to help maintain focus and direction
- **Resource allocation:** Effective utilization of resources including time, budget and personnel is critical. Proper resource allocation maximizes efficiency and enhances the likelihood of successful product development
- **Mitigating risks:** Identifying potential risks and obstacles throughout the product lifecycle allows teams to proactively manage challenges and adapt strategies as needed

- **Creating Market Differentiation:** Establishing a unique value proposition that distinguishes the product from competitors is vital. Differentiation helps attract customers and build brand royalty

Product Planning Tools and Techniques

S No	Technique	Description
1	SWOT Analysis	Evaluate product factors in terms of Strengths, Weakness, Opportunities and Threats
2	Market Segmentation	Identifies target audiences and customer personas to tailor marketing efforts
3	Competitor Analysis	Analyzes Competitors strengths, weaknesses and strategies to identify market positioning opportunities
4	Product Canvas	Outlines product features, benefits and the unique value proposition
5	Customer Journey Mapping	Visualizes customer interactions and pain points to enhance the user experience
6	Agile methodologies	Encourages iterative development and refinement of the product plan through sprints and feedback loops
7	Design thinking	Focuses on empathizing with customers, ideating solutions, and prototyping to meet user needs effectively
8	Business Model Canvas	Visualizes Business model components and their relationships for better strategic planning

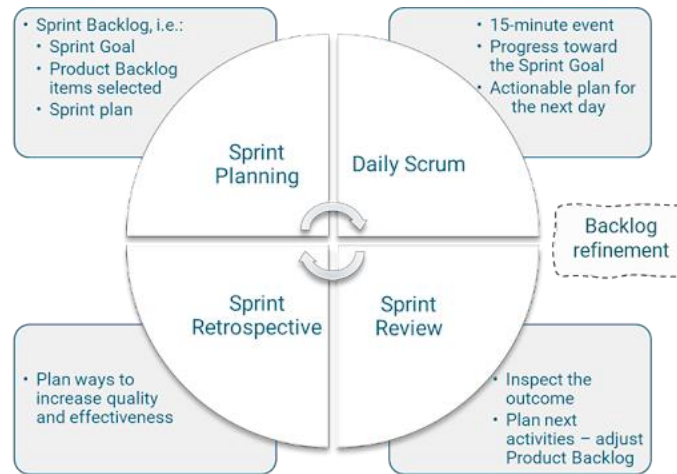
Agile Methodology

This is the prominent method used in many industries due to its simple and effective approach. This approach is based on **Sprints** – which are short term goals - for planning, monitoring and Reviewing of completion of specific tasks. The tasks are monitored daily/periodically, updated and continued again.

- **Sprint as a work period:** A short, focused period of time when a team works to achieve specific goals. Sprints can be any length, but are usually fairly short, such as one or two weeks.
- **Sprint planning:** The process where project teams determine which products or deliverables they can achieve within a set time frame (a sprint). The first Scrum meeting held at the start of every sprint.
- **Sprint review :** A meeting conducted at the end of every sprint where team members must discuss the work or tasks they have completed
- **Sprint retrospective:** A meeting that concludes the Sprint. It's held after a Sprint review and before the next Sprint starts

The sequence of steps adopted in the Agile methodology are illustrated in the Fig.

Problem Formation, Product Design and Specifications



Product Specifications

They outline the detailed requirements and characteristics of a product ensuring that it meets customer needs, industry standards and regulatory requirements. The various types of specifications that are followed are discussed here.

S No	Category	Functions
1	Functional specifications	Describe a product performance, features, and user experience. These specifications clarify what the product should do and how it should behave in different scenarios
2	Design specifications	Outline aesthetics, dimensions, materials, and user interface elements. They focus on how the product looks and feels, ensuring it meets the design standards and user preferences
3	Technical specifications	Detailed technical requirements, standards, testing criteria, and compatibility. They ensure that product integrates well with other systems and technologies
4	Quality specifications	Define quality standards, testing criteria, and reliability metrics. They provide benchmarks against which the product will be evaluated throughout its lifecycle
5	Regulatory specifications	Ensure compliance with laws, regulations, and industry standards for avoiding legal issues and ensuring product safety
6	Environmental specifications	Outline eco-friendliness, sustainability and waste management practices that focus on minimizing adverse environmental impact
7	Safety specifications	Define safety features, risk assessment processes and emergency procedures. They ensure the product is safe for consumers and complies with safety regulations

Sample specifications of a product are illustrated in the Fig.

BIKECHUNO KTM KTM RC 125				Engine : Honda Model : Dream 125 Type : 4-stroke, OHC Combustion : Direct injection, naturally aspirated Number of cylinder : 1 Bore : 52.4 mm Stroke : 57.9 mm Displacement : 124.9 cc Compression ratio : 9.3:1 Ignition system : CDI Ignition timing : 15° BTDC Cooling system : air cooled Sump capacity : 700 cc
SPECIFICATIONS				
Displacement 124.7 cc				
Power 14.75 bhp				
Transmission 6 Speed Manual	Fuel Type Petrol	Wheel Type Alloy	Ground Clearance 158 mm	
Fuel Tank Capacity 13.7 litres	Kerb Weight 162 kg	Wheelbase 1341 mm	Bore 58 mm	

Innovation towards Product Design

It refers to the process of developing new or improved products that meet evolving consumer needs, incorporate technological advancements, and enhance user experience. By embracing these approaches in product design, companies can differentiate themselves in a competitive market and respond effectively to customer demands.

S No	Key Element	Description
1	User-Centred Design	This approach prioritizes the needs and preferences of the end user throughout the design process. Techniques such as user research, persona development and usability testing are employed to gather insights and inform design decisions
2	Cross-Disciplinary Collaboration	Innovation in product design often involves collaboration between different disciplines such as engineering, marketing and design. This interdisciplinary approach fosters diverse perspectives and enhances creativity
3	Prototyping and Iteration	Rapid prototyping allows designers to create tangible representations of their ideas, facilitating testing and feedback. Iterative design processes enable teams to refine and improve products based on real world insights
4	Design Thinking	This model emphasizes empathy and creative problem-solving. By knowing the user's perspective and defining the problem clearly, teams can brainstorm innovative solutions and prototype potential designs
5	Sustainability considerations	Innovative product design increasingly focuses on sustainability by selecting eco-friendly materials thus minimizing waste

		production and making an eco-friendly product
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Stages in Product Design Innovation

S No	Stage	Description
1	Ideation	This stage involves brainstorming and generating wide range of ideas. Techniques such as mind mapping, sketching, and collaborative workshops can stimulate creative thinking
2	Concept Development	Selected ideas are further refined into viable concepts. Designers create sketches, digital models, or initial prototypes to visualize the product and assess its feasibility
3	Prototyping	Creating prototypes allows designers to test and validate their concepts using either low-fidelity or high-fidelity prototypes to gather feedback from users.
4	Testing and validation	Prototypes are subjected to user testing to gather insights into usability and performance. Feedback is analyzed to identify areas for improvement and ensure the product meets user needs
5	Final Design & Development	Once the design is validated, detailed specifications are created including materials, dimensions, and manufacturing processes. This stage also involves collaboration with engineering teams to ensure the product is easily manufactured
6	Market launch	After finalizing the design and production processes, the product is launched into the market. Marketing strategies are developed to promote the new product and reach the target audience effectively
7	Post-launch Evaluation	Following the product launch, businesses should continuously monitor performance, gather customer feedback, and assess market reception, which is essential for future improvements

Challenges in Product Design Innovation

S No	Challenge	Description
1	Resource Constraints	Limited budgets and time constraints can hinder the innovation process. Companies must allocate resources effectively to balance creativity and with practicality
2	Market Uncertainty	Predicting market trends and consumer preferences can be challenging. Businesses must remain agile and adaptable to respond to changing conditions
3	Balancing Innovation and Functionality	Striking balance between innovative features and product usability is critical. Highly complex designs can alienate users, while very simplistic designs may not meet needs
4	Managing Cross-Disciplinary teams	Collaborating with diverse teams can lead to communication challenges. Clear project management and defined roles are essential for ensuring alignment and productivity
5	Compliance and	Products must adhere to various regulations and industry standards,

	Regulation	which can complicate the design process. Understanding and incorporating these requirements early on is vital
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CASE STUDIES OF IMPROVED PRODUCT DESIGNS

1. Heinz Ketchup Bottle

Problem with existing Model:

1. Difficulty in getting the last drop of ketchup out of bottle due to its thick consistency
2. Difficulty in handling and squeezing the bottle with one hand
3. Spread of ketchup along the walls of bottle due to squeezing



Improvement

1. Inverted bottle design making easy to squeeze
2. No spaying and wasting of the ketchup
3. Easy handgrip and no slipping
4. Easy to empty the contents

Outcome

1. Easily handled with single hand
2. Reduced wastage
3. Good appearance
4. Increased user satisfaction
5. Unbreakable material used
6. No risk of damage when children are using

2. Music Player vs.iPoD Shuffle

Problem with existing portable music player:

1. Highly complex features making user confused
2. Difficulty in carrying and securing the device during active working
3. Hard to operate

Improvement

1. iPod shuffle has sleek and simple design
2. Essential controls for music playback
3. Can be easily attached to a personal wear
4. Reduced complexity, more focus of functioning buttons

Outcome

1. Reduced unnecessary features
2. Portable and can be easily carried

Problem Formation, Product Design and Specifications

3. Easy-to use options
4. Sleek design occupying reduced space

		
Music Player with Blue-toothconnection (Original)	iPoD Shuffle – Sleek Design (Improved design)	iPoD Shuffle – 6G with touch screen options (updated design)